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Abstract

In this coursework, the new encryption approach invented in this coursework uses a classical substitution strategy in conjunction with the use of the bitwise logical operations to enhance the protection of the data. The paper starts with the examination of the Caesar Cipher, including its simple encryption and decryption mechanism along with the limitations, the most notable weaknesses being the vulnerability to brute force and frequency-based attacks. The solution to this shortcoming is a hybrid cryptography scheme that is called Dynamic Layered Caesar XOR (DLC-XOR).

The algorithm uses position-dependent Caesar shifting, text reversal and XOR based encryption to add more unpredictability and diffusion to the ciphertext. The mathematical concepts that have been used under it such as modular arithmetic and reversible XOR logic are explained in a step-by-step manner.

The use of more than one plaintext input can prove that the proposed algorithm is always able to provide correct encryption and decryption. Even though the scheme is mainly used in the educational and experimental context, it demonstrates that the classic methods of cryptography may be improved using the stratified design and logical operations and provides practical experience of making both algorithms and principles of data security.

1. Introduction

1.1 Understanding Security

In this coursework, security is approached as a practical requirement rather than a theoretical concept. The focus is placed on how information can be protected during storage and transmission, particularly when simple encryption techniques are applied to reduce unauthorized access. In the framework of cybersecurity and information systems, information security concerns the protection of data, network, and applications against numerous threats, including hacking attacks, data breach, malware, and physical theft. Security is implemented using a mix of preventive controls like encryption, firewalls, authentication features and access controls and corrective and recovery controls like data backups, incident response processes and disaster recovery processes. A security framework in place well assists organizations to detect vulnerabilities, deal with risks, and make the information systems reliable and trustworthy (Barney & Lutkevich, 2022).



Figure 1: Security in Cryptographic Systems

1.2 CIA Triad in information Security

The CIA triad is used as a reference model to evaluate whether the proposed encryption method maintains confidentiality, preserves data integrity after encryption and decryption, and allows authorized access to information when required. An effective system satisfies all three components: confidentiality, integrity, and availability. An information security system that is lacking in one of the three aspects of the CIA triad is insufficient.

The CIA Triad is a fundamental model in information security that rests on the following:

- **Confidentiality:** To ensure that sensitive information is not compromised through unauthorized access by employing access control mechanisms, encryption methods and secure authentication processes.
- **Integrity:** Quality and authenticity of the data are maintained by integrity methods such as hashing and error-checking that help to prevent the data from being tampered with.
- **Availability:** The information systems and data should be available whenever needed by the authorized users, and it is achieved through redundancy, frequent backups and defence against denial-of-service attacks (Fortinet.com, 2024).



Figure 2: CIA Triad

1.3 Cryptography: An Introduction

In this project, cryptography is treated as a functional mechanism for transforming readable text into protected formats. The emphasis is not on complex algorithms but on demonstrating how combining classical substitution techniques with logical operations can enhance resistance against basic attacks. Cryptographic keys and algorithms are used in encrypting the data. Plaintext is the original data which is encrypted to form ciphertext and the only way it can be revived is through the decryption process with the proper key. In information systems, cryptography is an important tool in attaining confidentiality and integrity. Contemporary cryptography is based on mathematical methods and algorithm complexity to make sure that the data encrypted cannot be easily decrypted (Khan Academy, 2024).

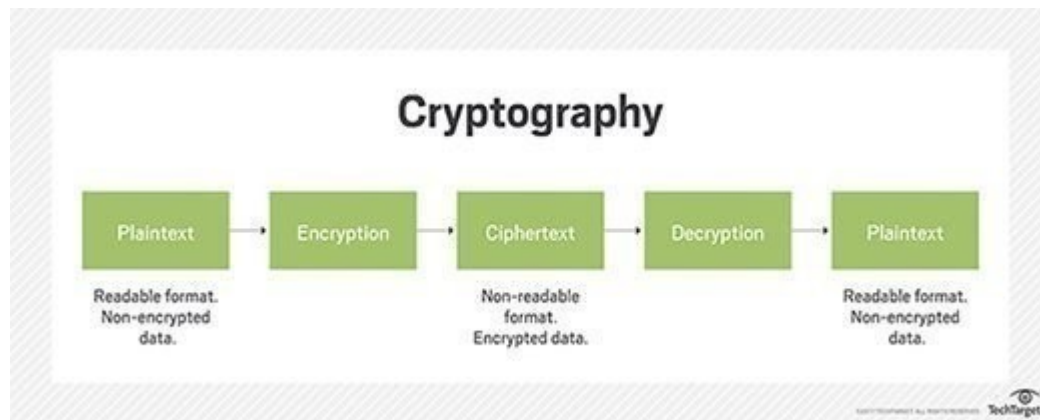


Figure 3: Cryptography

1.4 Key Terminologies in Cryptography

1.4.1 Plaintext and Ciphertext

- Plaintext: Plaintext is the original readable information before encrypting it.

- Ciphertext: Ciphertext is scrambled and unreadable information after encrypting the information to ensure that it is not accessed by an unauthorized party (cloudflare, 2024).

1.4.2 Encryption and Decryption

- Encryption: Encryption refers to the act of transforming plaintext into ciphertext with the help of a cryptographic algorithm and a key.
- Decryption: Decryption refers to the process of returning the ciphertext to the original plaintext that can be understood by authorized parties (Khan Academy, 2024).

1.4.3 Keys

In cryptography, a key is a string of characters used within an encryption algorithm for altering data so that it appears random. Like a physical key, it locks (encrypts) data so that only someone with the right key can unlock (decrypt) it (cloudflare, 2024).

1.4.4 Hashing

Hashing refers to the process of generating a fixed-size output from an input of variable size using the mathematical formulas known as hash functions. This technique determines an index or location for the storage of an item in a data structure (GeeksforGeeks, 2024).

1.4.5 Digital Signatures and Certificates

Digital signatures and digital certificates are two fundamental technologies used to ensure security, authenticity, and trust in online communication. They are widely used in areas such as online banking, secure emails, software distribution, e-commerce, and electronic document signing (GeeksforGeeks, 2024).

1.5 History of Cryptography

1.5.1 Early Cryptographic method

The first cryptographic techniques were essentially simple substitution and transposition systems aimed at concealing the meaning of messages to non-authorized readers. The Caesar Cipher is among the oldest of this sort; the letters in a message were moved through the alphabet by a fixed number of positions. These techniques were convenient to use and worked in their time especially in military and diplomatic communication. The initial cryptographic systems were, however, based on secrecy and not mathematical

strength, and they were therefore susceptible to attack as soon as their logic was uncovered (Encyclopedia Britannica , 2024).



Figure 4: Cryptography Methods

1.5.2 Evolution During the World Wars

Cryptography developed at a significantly faster pace during the world wars due to the urgent need for secure military communication. Complex key-based encryption with rotating parts was first introduced with mechanical and electromechanical encryption machines, like an Enigma machine of the German military in World War II. These systems were a significant breakthrough of manual methods to machine-aided cryptography. The successful cracking of Enigma by Allied cryptographers also caused the value of designing algorithms well and conducting stringent security testing to be emphasized over the use of obscurity (Singh, S., 1999).



Figure 5: The Enigma Machine

1.5.3 Modern Cryptography

Cryptographic techniques have evolved over time from simple manual ciphers to complex mathematical algorithms. However, this coursework does not aim to analyze historical developments in detail. In comparison to the classical techniques, the modern cryptographic algorithms are based on complexity and computational hardness of the mathematical to ensure security. The widespread use of symmetric algorithms like AES

and asymmetric algorithms like RSA is to protect the digital data, online transactions and communication protocols. The other enhanced security features included in modern cryptography are digital signatures, authentication, and secure exchange of keys, and as such, cryptography has become an essential part of the current cyber security system (NIST, 2023).

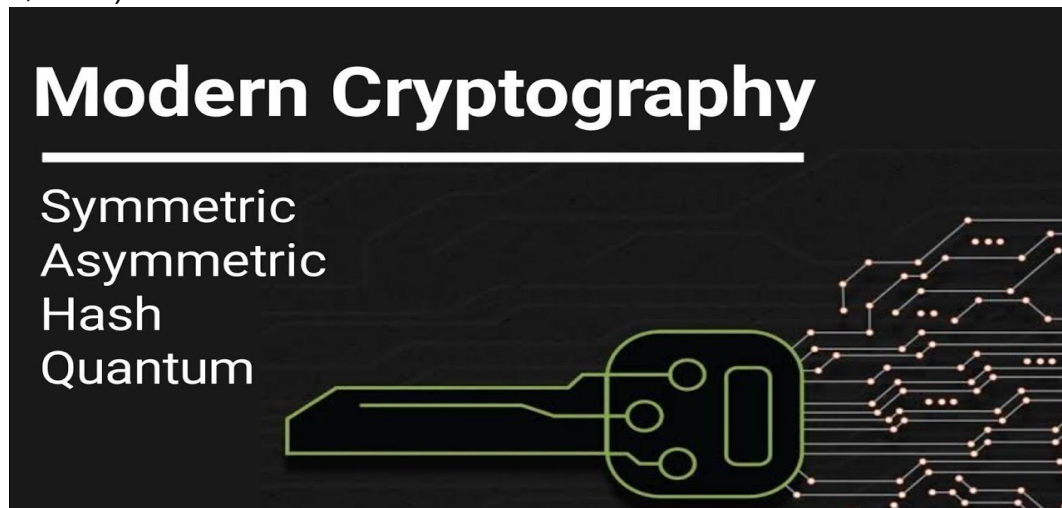


Figure 6: Modern Cryptography

1.6 Symmetric and Asymmetric Cryptography

1.6.1 Symmetric Cryptography

Symmetric Key Cryptography, also known as conventional or single-key encryption was the main means of encryption prior to the birth of the public key cryptography in the 1970s. In symmetric-key algorithms, data encryption and decryption use the same keys. This form is cryptography important to the security of data since the same key is employed in encryption and the key is employed to decrypt (GeeksforGeeks, 2025).

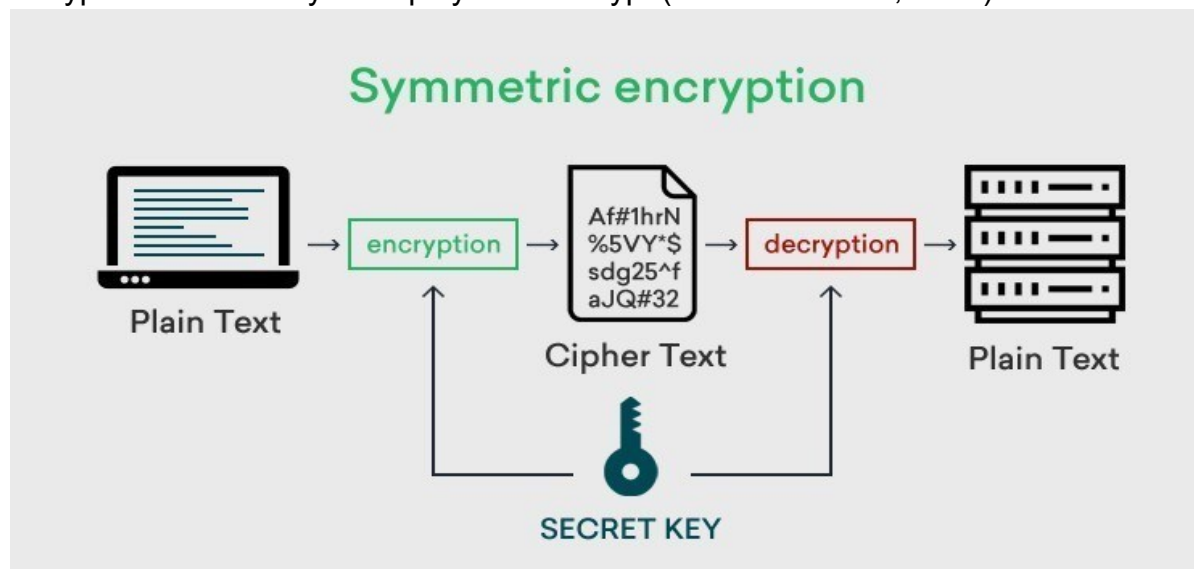


Figure 7: Symmetric encryption

1.6.2 Asymmetric Cryptography

In asymmetric Key cryptography, two keys are used which are also referred to as key pairs; one is the public key and the other is the private key. The public key is made publicly available. Anybody can encrypt messages using this public key, and only the receiver has access to the corresponding private key, which can decrypt such messages. Another form used to represent Asymmetric Key cryptography is known as public-key cryptography (GeeksforGeeks , 2025).

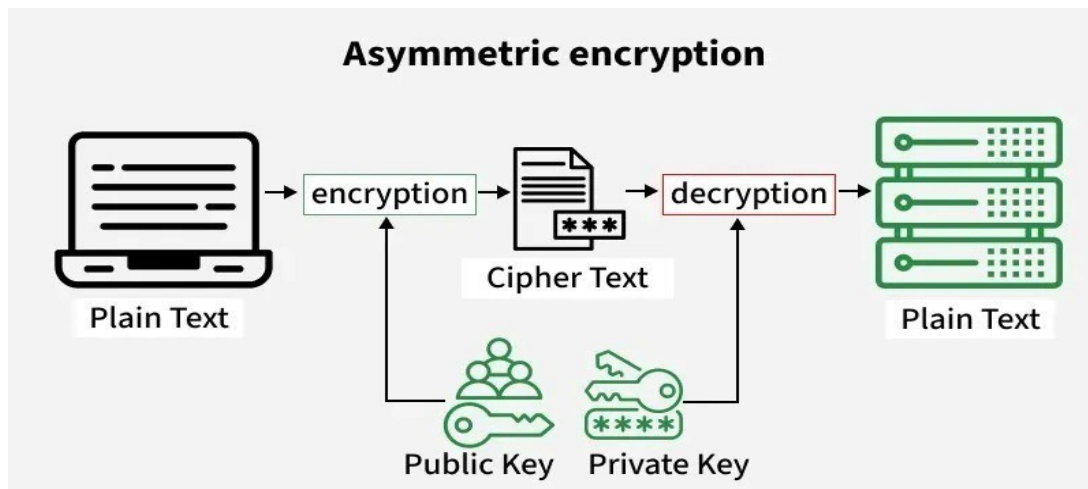


Figure 8: Asymmetric encryption

1.6.3 Key Differences

Encryption is one of the most basic concepts in the world of cybersecurity as it ensures that some information does not fall into the wrong hands. There are two primary types of encryption techniques, namely, symmetric key encryption and asymmetric key encryption. Anyone involved in data security must know the differences between these two methods.

- Symmetric Key: It only requires a single key for both encryption and decryption.
- Asymmetric Key: It requires two keys, a public key and a private key, one to encrypt and the other to decrypt.

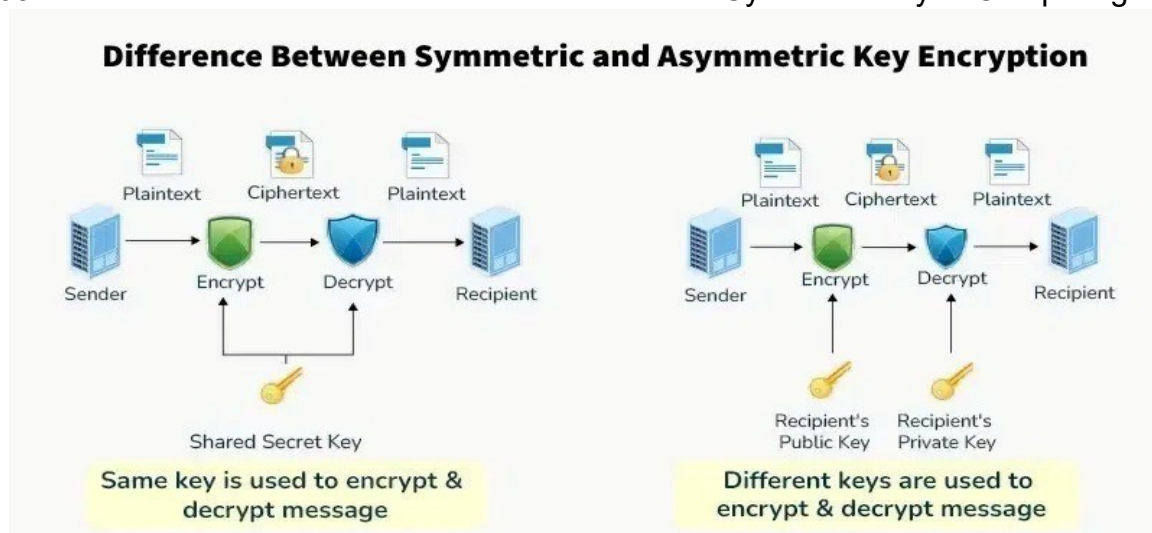


Figure 9: Difference Between Symmetric and Asymmetric

2. Selection of Cryptographic Algorithm

The Caesar Cipher is selected in this coursework as one of the simplistic symmetric key encryption techniques where each letter is moved by a predetermined number of places. It can be easily used to study the basics of such concepts as plaintext, ciphertext, keys, encryption, and decryption. Though not safe according to the modern standards, it offers a clear base to build and improve a new cryptography system by using logical and mathematical enhancements.

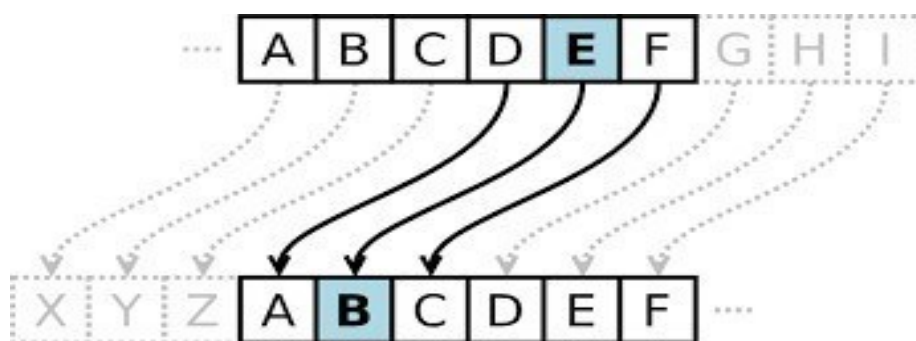


Figure 10: The Caesar Cipher

2.1 Background

The Caesar Cipher is historically associated with Julius Caesar and represents one of the earliest substitution-based encryption techniques, where characters are shifted by a fixed number of positions within the alphabet. The algorithm operates by substituting every character in the plaintext by the corresponding character which is several places farther down the alphabet. In the given example, the shift value is three and in trying to change the letter A to D, B to E, etc.

This cipher is based upon one secret key which is the shift value. The receiver and the sender are required to be aware of this key so that they can encrypt and decrypt the message. The common element makes Caesar Cipher to be categorized as a symmetric encryption algorithm (Britannica , 2024).

2.2 Why Caesar Cipher?

- Easy and comprehensible- perfect to know the basic concepts of cryptography.
- Historical importance- one of the oldest known encryption techniques.
- Extendable - is capable of being extended with other logic or mathematical operations.
- Educational value - is a clear representation of symmetric key encryption.

2.3 Working Principle of the Caesar Cipher

1. Encryption Formula:

$$C = (P + K) \bmod 26$$

Where :

C = Position of ciphertext letter (0-25)

P = Plaintext letter position (0–25)

K = Shift/key value

26 = Number of letters in the English alphabet

2. Decryption Formula:

$$P = (C - K) \bmod 26$$

3. Case Sensitivity:

- Historically, Caesar Cipher takes upper case letters only, ignoring lower case and non-alphabetic characters.
- Modern versions can deal with capital and lower-case letters.

4. Key Range:

- A 26 letter English alphabet contains 1-25 range of K.
- K = 0 or K = 26 leaves the plaintext unchanged.

5. Symmetric Nature:

- Both encryption and decryption are done using the same key K, thus it is a symmetric key algorithm.

2.4 Encryption and Decryption Examples

For Encryption

In the Caesar Cipher, the actual encryption is performed by rotating every letter of the plaintext through the alphabet by a given number of slots. This constant figure is referred to as the shift key.

$C = (P + K) \bmod 26$ where C is the numerical value of the Cipher Text, P is the numerical value of the Plain Text and K is the Key.

For Decryption

The ciphertext is decrypted through the reciprocation of each ciphertext letter by the identical key to revert to the original plaintext.

$P = (C - K) \bmod 26$ where C is the numerical value of the Cipher Text, P is the plaintext and K is the key.

For Ceaser Cipher, the value of shift will be by 3 positions.

Plain	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

Table 1: Encryption and Decryption Examples

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Table 2: Numbering of Encryption and Decryption Examples

Now, encrypting the plain text ABINASH as follows:

$$A \rightarrow C = (P + K) \bmod 26 = (0+3) \bmod 26 = 3 = D \quad B \rightarrow$$

$$C = (P + K) \bmod 26 = (1+3) \bmod 26 = 4 = E$$

$$I \rightarrow C = (P + K) \bmod 26 = (8+3) \bmod 26 = 11 = L$$

$$N \rightarrow C = (P + K) \bmod 26 = (13+3) \bmod 26 = 16 = Q$$

$$A \rightarrow C = (P + K) \bmod 26 = (0+3) \bmod 26 = 3 = D$$

$$S \rightarrow C = (P + K) \bmod 26 = (18+3) \bmod 26 = 21 = V$$

$$H \rightarrow C = (P + K) \bmod 26 = (7+3) \bmod 26 = 10 = K$$

Hence, the Cipher Text is DELQDVK

Again, Decrypting the cipher text “ DELQDVK”

$$D \rightarrow P = (C - K) \bmod 26 = (3-3) \bmod 26 = 0 = A$$

$$E \rightarrow P = (C - K) \bmod 26 = (4-3) \bmod 26 = 1 = B$$

$$L \rightarrow P = (C - K) \bmod 26 = (11-3) \bmod 26 = 8 = I$$

$$Q \rightarrow P = (C - K) \bmod 26 = (16-3) \bmod 26 = 13 = N$$

$$D \rightarrow P = (C - K) \bmod 26 = (3-3) \bmod 26 = 0 = A$$

$$V \rightarrow P = (C - K) \bmod 26 = (21-3) \bmod 26 = 18 = S$$

$$K \rightarrow P = (C - K) \bmod 26 = (10-3) \bmod 26 = 7 = H$$

Hence, the plain Text is ABINASH.

2.5 Advantage and Disadvantages of Caesar Cipher

2.5.1 Advantages

The Caesar Cipher is also easy and simple to learn and hence it is best used in learning the basics of cryptography including plaintext and ciphertext, keys, encryption and decryption. It is computationally inexpensive, and it consumes very little resources because it uses just simple modular arithmetic. The cipher is also useful in explaining how secure communication systems were developed early because it is historically significant. It is also easy to extend or mix with other methods because of its simple design, which is why it is preferable as a foundation model of experimentally improving cryptography.

2.5.2 Disadvantage

The key space of the Caesar Cipher is very small and therefore the algorithm is extremely susceptible to the brute-force attack. It also retains the pattern of letters frequency such that attackers can easily decode the cipher by use of frequency analysis. The fixed value of the shift makes the encryption patterns predictable, and original Caesar cipher does not support numbers or special characters. which limits its practical application. The above weaknesses render Caesar Cipher inappropriate in contemporary safe communication and necessitate the more advanced methodologies like the Dynamic Layered Caesar XOR (DLC-XOR) algorithm.

3. Development

This section relates the design and development of a new proposed cryptographic method called the Dynamic Layered Caesar XOR (DLC-XOR). The algorithm is designed to overcome the inherent drawbacks of the classical Caesar cipher since it uses dynamic shifting, multi-layered encryption and XOR-based logical operations.

The main point of this development is to show how successful the means of traditional and simple cryptography can be reinforced with structure and mathematics as well as logic. Although the classical Caesar cipher uses only one fixed substitution rule, the DLCXOR algorithm uses several transformation steps to break predictable patterns in the plaintext thus helping to enhance immunity to simple cryptanalysis strategies.

Although the algorithm is inspired by the classical Caesar cipher, the incorporation of dynamic shifting and XOR-based bitwise operations enhances its cryptographic strength.

3.1 Name of the Algorithm

Dynamic Layered Caesar XOR (DLC-XOR)

3.2 Research on the Caesar Cipher and Its Modifications

One of the oldest known methods of encryption, named after Julius Caesar, is the Caesar Cipher, in which the letters of the plaintext are moved by some fixed number of places in the alphabet. The cipher is deemed to be highly insecure in modern times despite its historical significance because it has limited key space, is linearly substituted, and has a predictable structure.

Cryptographic analysis today indicates that the Caesar cipher is vulnerable to easy breaking through brute-force attacks or frequency analysis and can be broken in seconds in most cases (Stallings, 2018). Various improvements have been suggested in cryptographic literature and in scholarships to avoid these shortcomings:

- **Dynamic Shift Caesar Cipher:** Depending on the location of each character, the shift value is altered, making it less predictable and making it harder to perform a frequency analysis.
- **Keyed Caesar Cipher:** The magnitude of the shift is defined by a secret key which makes it harder to decrypt without authorization.
- **Layered Caesar Cipher:** Several cycles of Caesar rotation have been used and may be combined with logic operations like inverting the text.
- **XOR-Based Enhancement:** Applications of the XOR (Exclusive OR) operation in modern cryptography XOR (Exclusive OR) operations are frequently utilized in contemporary cryptographic techniques because of their reversibility and the capacity to obscure statistical distribution at the bit level.

3.3 Mathematical and Logical Foundations of DLC-XOR

The algorithm offered has a foundation on established mathematical and logical principles that are widely applied in designing cryptographic systems.

1) Modular Arithmetic

Modular arithmetic makes sure that the character shifts do not fall out of the valid alphabet range (025). This is expressed as:

$$C1 = (P + K + i) \bmod 26$$

Where:

- P denotes the index of the plaintext character.
- K is the base encryption key.
- i is the position index of the character.
- C1 is the mid-way encrypted text.

2) Dynamic Shift Mechanism

The DLC-XOR algorithm uses a position-dependent shift as opposed to the classical Caesar cipher where the shift is a fixed value. This makes sure that even the same characters that are displayed at varying positions give different encrypted results and this greatly increases unpredictability.

3) XOR Logical Operation

The XOR (Exclusive OR) is an operation of the ASCII values of characters that is a reversible bitwise logical operation. Its cryptographic strengths are:

- Reversible: When XOR is applied twice using the same key, the initial value is recovered.
- Pattern masking: XOR causes non-linearity, which interferes with statistical patterns.
- Lasting support: It allows the encryption of non-alphabetic characters.

XOR is a bit-level encryption, unlike substitution ciphers, which has enhanced resistance to cryptanalysis based on frequency.

4) Text Reversal

Between layers of encryption, text reversal is used to disorient positional alignment of plaintext and ciphertext. This rational alteration also makes cryptanalysis more difficult by disorienting relationships in order of character sequences.

5) Secondary Encryption Layer

Upon reversal, an extra encryption layer is applied with the help of the XOR and keybased manipulation: $C2 = C1 \oplus \text{XOR_Key}$ Where:

C1 = Character ASCII value

XOR_Key = XOR encryption key

C2 = final Ciphertext value

This stratification system improves diffusion, and it reinforces pattern distortion throughout the ciphertext.

3.4 Creation of the DLC-XOR Algorithm

DLC-XOR algorithm is a hybrid algorithm that encrypts by using substitution, logical encryption methods to ensure that alphabetic and non-alphabetic data get ciphered.

3.4.1 Alphabet Indexing Table

The alphabet indexing table maps each alphabetic character to its corresponding numerical index (0-25). In the DLC-XOR algorithm, the dynamic shift value does not depend on the alphabet itself but instead depends on the base key (K) and the position index (i) of each character in the plaintext.

The shift applied during encryption is calculated dynamically using the formula:

$$\text{Dynamic Shift} = K + i$$

Therefore, the shift value varies according to the character's position in the plaintext, ensuring that identical characters occurring at different positions produce different ciphertext values.

Alphabet	Index	Shift Value
A	0	3
B	1	4
C	2	5
D	3	6
E	4	7
F	5	8
G	6	9
H	7	10
I	8	11
J	9	12
K	10	13
L	11	14
M	12	15
N	13	16
O	14	17
P	15	18

Q	16	19
R	17	20
S	18	21
T	19	22
U	20	23
V	21	24
W	22	25
X	23	0
Y	24	1
Z	25	2

Table 3: Indexing of Alphabets and Key Values

3.4.2 New Encryption Algorithm

Plaintext: ABINASH

Key: Shift by 3

XOR Key = 5

Layer 1: Dynamic Caesar Shift

Position (i)	Plaintext	Index (p)	Calculation	K+i	Shift Index(C1)	Cipher (C1)	C1 ASCII	C1 Binary (8-bit)
0	A	0	$(3 + 0) \bmod 26$	$3 + 0 = 3$	3	D	68	01000100
1	B	1	$(1 + 4) \bmod 26$	$3 + 1 = 4$	5	F	70	01000110
2	I	8	$(8 + 5) \bmod 26$	$3 + 2 = 5$	13	N	78	01001110
3	N	13	$(13 + 6) \bmod 26$	$3 + 3 = 6$	19	T	84	01010100

4	A	0	$(0 + 7) \bmod 26$	$3 + 4 = 7$	7	H	72	01001000
5	S	18	$(18 + 8) \bmod 26$	$3 + 5 = 8$	$26 \bmod 26 = 0$	A	65	01000001
6	H	7	$(7 + 9) \bmod 26$	$3 + 6 = 9$	16	Q	81	01010001

Table 4: New Dynamic Caesar Shift

The encryption is of a position-dependent offset based on the formula:

$$C1 = (P + K + i) \bmod 26$$

Where:

- P = plaintext index at position i
- K = base key
- i = position index at position
- C1 = intermediate ciphertext index at position

Output Layer 1: C1 = DFNTHAQ

3.4.3 Text Reversal

The middle ciphertext is reversed to bring about positional confusion. The only difference is that the order of characters is reversed; the values of ASCII and binary are identical.

Text Reversal Formula

- $C1 = \{C0, C1, C2, \dots, Cn-1\}$ be the intermediate ciphertext
- n be the length of the text.

The reversed ciphertext C1 is computed as:

$$C1(i) = C1(n-1-i)$$

Where:

$$i = 0, 1, 2, \dots, n - 1$$

C1(i) is the character at position i after reversal

Reversed Position	Original Position	Character	ASCII	Binary (8-bit)
-------------------	-------------------	-----------	-------	----------------

0	6	Q	81	01010001
1	5	A	65	01000001
2	4	H	72	01001000
3	3	T	84	01010100
4	2	N	78	01001110
5	1	F	70	01000110
6	0	D	68	01000100

Table 5: Text Reversal

Output After Text Reversal

C1 = QAHTNFD

3.5 Secondary Encryption Layer

The DLC-XOR algorithm is then used to encrypt a second layer after text reversal with a second encryption layer based on a bitwise XOR operation. It is at this stage that the reversed intermediate ciphertext is converted at the binary level to give the final encrypted characters.

XOR Encryption Formula

$$C2 = (C1 \oplus \text{XOR_Key}) \bmod 256$$

XOR is a reversible operation, applying the same XOR key during decryption restores the original ASCII values, ensuring correct plaintext recovery.

Where:

- C1 is the ASCII of the inverted value of the middle-encrypted character.
- XOR_Key represents the XOR encryption key (5).
- $\oplus$ denotes the XOR operation
- The last encrypted ASCII character is C2.

Input to XOR Layer

Reversed intermediate Ciphertext:

QAHTNFD

ABINASH KUMAR SAH

Character	ASCII Values	Binary (8-bit)
Q	81	01010001
A	65	01000001
H	72	01001000
T	84	01010100
N	78	01001110
F	70	01000110
D	68	01000100

Table 6: Reversed intermediate ciphertext

Detailed XOR Computation Table

The XOR operation is applied bit-by-bit using the XOR key 00000101.

Character	ASCII(C1)	ASCII (Binary)	XOR Key (Binary)	XOR Result (Binary)	Result (Decimal)	Cipher
Q	81	01010001	00000101	01010100	84	T
A	65	01000001	00000101	01000100	68	D

H	72	01001000	00000101	01001101	77	M
T	84	01010100	00000101	01010001	81	Q
N	78	01001110	00000101	01001011	75	K
F	70	01000110	00000101	01000011	67	C
D	68	01000100	00000101	01000001	65	A

Table 7: Detailed XOR operation

Final XOR Output: TDMQKCA

The TDMQKCA final ciphertext is obtained after XOR encryption. In decryption, the XOR operation is reversed with the same XOR key and the text in the middle is then reversed again to get the original casing order and then the reverse dynamic Caesar shift is applied.

Reverse Dynamic Caesar Shift (Decryption Layer)

The plaintext is recovered using:

$$P = (C1 - K - i) \bmod 26$$

Position(i)	C1 Character	C1 Index	Reverse Shift	Plaintext Index	Plaintext
0	D	3	3	0	A
1	F	5	4	1	B

2	N	13	5	8	I
3	T	19	6	13	N
4	H	7	7	0	A
5	A	0	8	18	S
6	Q	16	9	7	H

Table 8: Reversed Dynamic Caesar shift

Plaintext: ABINASH

3.5.1 Elaboration of the New Algorithm

Dynamic Layered Caesar XOR (DLC-XOR) algorithm is an improvement of the classical Caesar cipher algorithm that adds dynamically changing shifts, encrypting at layers as well as logical XOR operations which makes the algorithm much stronger. The aim of this augmentation is to minimize predictability, enhance complexity as well as augment resistance to simple cryptanalytic attacks.

3.5.2 Why the modifications Were Necessary

Classical Caesar cipher has a fixed shift value, and it is easy to crack it by brute-force and frequency analysis. It also displays only the alphabetic characters and gives no security to the numeric or other special characters. These constraints render the initial cipher unsuitable in the current security requirements.

To overcome these weaknesses, the DLC-XOR algorithm proposes dynamic shifting in terms of character position, multi-level encryption and XOR-based logical operations. These changes were required to lessen the repetition of patterns, raise the variation of the keys and to mask the data at the bit level

3.5.3 The New Methodology

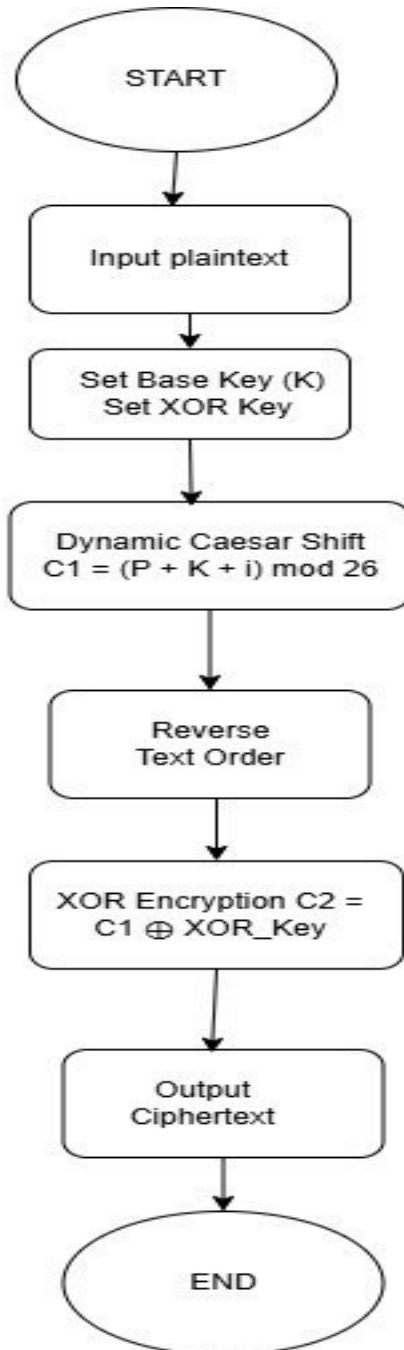
There are three steps of encryption used in the DLC-XOR methodology. In the first step, an active Caesar shift is done, with a base key and position of the character. Second, positional relationships are broken by reversing the text that results. Lastly, XOR encryption of ASCII values of the characters is done to secure on a bit level.

The use of decryption is the opposite process, and it ensures proper recovery of the original plaintext. The hierarchical and logical approach enhances the level of security compared to classical ciphers and is at the same time easy and appropriate in learning.

3.6 Flowchart of the New Algorithm

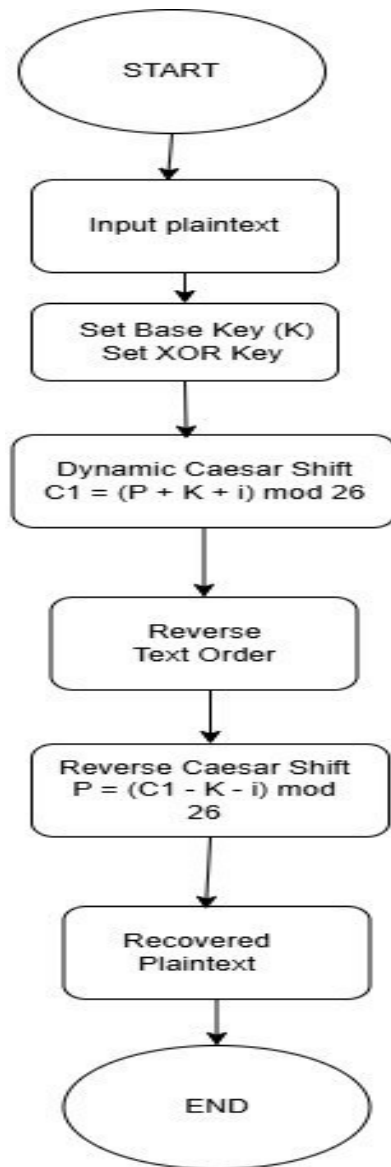
The flowchart includes a decision stage that determines whether the input character is alphabetic or non-alphabetic, ensuring the appropriate encryption path is selected.

For Encryption



For Decryption

The decryption flowchart reverses these steps to accurately reconstruct the original plaintext.



4. Testing

In this section, the entire DLC-XOR encryption stages are used to test the clarity of the tests, and the dynamic Caesar shifting, text reversal and XOR encryption are performed on alphabetic characters. Numbers and special characters go straight through the Caesar and reversal layers and are encrypted directly with ASCII-based XOR operation. This design will provide proper manipulation of any type of character without compromising the reversibility of the algorithm.

Note: All tests are character-based, as the algorithm is a variation of the classical Caesar Cipher, which operates on individual characters.

4.1 Test 1: ALPHABET

For the encryption,

Plaintext = ALPHABET

- Key: Shift by 4
- XOR Key:6

Plaintext	Index(P)	Shifted Index ($C=(P+K) \bmod 26$)	Ciphertext	ASCII	XOR key	XOR Output (Decimal)	XOR Ciphertext
A	0	4	E	69	6	79	O
L	11	15	P	80	6	86	V
P	15	19	T	84	6	90	Z
H	7	11	L	76	6	82	R
A	0	4	E	69	6	79	O
B	1	5	F	70	6	76	L

E	4	8	I	73	6	79	O
T	19	23	X	88	6	94	^

Table 9: Encryption of Test 1

Ciphertext: OVZROLO^

For the decryption,

- Ciphertext: OVZROLO^
- Key: Shift by 4
- XOR Key: 6

XOR Ciphertext	XOR Output	ASCII	Shifted Index (C)	Plaintext
O	79	69	4	A
V	86	80	15	L
Z	90	84	19	P
R	82	76	11	H
O	79	69	4	A
L	76	70	5	B
O	79	73	8	E
^	94	88	23	T

Table 10:Decryption of Test 1

Plaintext: ALPHABET

TEST NO	1
OBJECTIVE	To verify correct encryption and decryption of a longer plaintext using the Modified Caesar Cipher with XOR.
ACTION	The plaintext ALPHABET is encrypted using a Caesar shift key of 4, followed by XOR encryption using key 6. The resulting ciphertext is then decrypted by applying XOR decryption and reverse Caesar shifting.
CONCLUSION	The original plaintext is successfully reconstructed, confirming that the algorithm functions correctly for longer character sequences.

4.2 Test 2: Numeric Encryption

For the encryption,

Plaintext: 123456

- Key: Shift by 3
- XOR key: 5

Plaintext	Ciphertext	ASCII	XOR Key	XOR Output	XOR Ciphertext
1	4	49	5	52	4
2	5	50	5	55	7
3	6	51	5	54	6
4	7	52	5	49	1
5	8	53	5	48	0
6	9	54	5	51	3

Table 11: Encryption of Test 2

Ciphertext: 476103

For the Decryption,

- Ciphertext: 476103
- Key: Shift by 3
- XOR key: 5

XOR Ciphertext	XOR Output	ASCII	Shifted Index	Plaintext
4	52	49	4	1
7	55	50	5	2
6	54	51	6	3
1	49	52	7	4
0	48	53	8	5
3	51	54	9	6

Table 12: Decryption of Test 2

Plaintext: 123456

Test	2
Objective	To test the encryption and decryption of a numeric-only plaintext.

Action	Through the ASCII based XOR encryption, the numeric plaintext 123456 is encrypted directly because the numeric characters bypass the Caesar shift layer. XOR decryption is used in decryption to reconstruct the original plaintext.
Conclusion	The original numeric plaintext 123456 is accurately recovered, demonstrating the algorithm's effectiveness with numeric data.

4.3 Test 3: Alphanumeric Value

- Plaintext: AS123
- Shift Key: 2
- XOR Key: 7
- **Encryption Table**
 - Numeric characters bypass the Caesar shifting stage and are encrypted directly using ASCII-based XOR operations.

Plaintext	Type	Index(P)	Shifted Index ($C=(P+K)\bmod 26$)	Ciphertext	ASCII	XOR Key	XOR Output	XOR Ciphertext
A	Alpha	0	2	C	67	7	68	D
S	Alpha	18	20	U	85	7	82	R
1	Num	-	-	3	49	7	54	6
2	Num	-	-	4	50	7	53	5

3	Num	-	-	5	51	7	52	4
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Table 13: Encryption of Test 3

Ciphertext: DR654

For the decryption,

- Ciphertext: DR654
- Key: shift by 2
- XOR key: 7

XOR Ciphertext	XOR Output	ASCII	Shifted Index	Plaintext
D	67	68	2	A
R	82	84	20	S
6	49	54	-	1
5	50	53	-	2
4	51	52	-	3

Table 14: Decryption of Test 3

Plaintext: AS123

Test	3
Objective	To verify correct encryption and decryption of an alphanumeric plaintext.

Action	The alphanumeric plaintext AS123 is alphabetic characters are encrypted using a Caesar shift key of 2, while numeric characters bypass the Caesar layer and are encrypted using XOR.
Conclusion	The original plaintext AS123 is successfully reconstructed, showing the algorithm handles mixed alphabetic and numeric characters reliably.

4.4 Test 4: Special Characters

For the encryption,

- Plaintext: @#&*
- XOR Key: 3

Plaintext	ASCII	XOR Key	XOR Output	Ciphertext
@	64	3	67	C
#	35	3	32	_
&	38	3	37	%
*	42	3	41	)

Table 15: Encryption of test 4

Ciphertext: C _%)

For the decryption,

- Ciphertext: C _%)
- Key: Shift by 1
- XOR key: 3

XOR Ciphertext	XOR Output	ASCII	Plaintext
C	64	64	@
_	35	35	#
%	38	38	&
)	42	42	*

Table 16: Decryption of Test 4

Test	4
Objective	To test DLC-XOR encryption and decryption for special characters and symbols.

Action	The plaintext @#&* is encrypted using ASCII-based XOR encryption with key 3. Since special characters cannot be shifted like letters, the Caesar shift is not applied. Decryption is performed using XOR decryption to recover the original plaintext.
Conclusion	The original plaintext @#&* is accurately recovered, confirming the algorithm can process symbols and punctuation correctly.

4.5 Test 5: Mixed Case with Numbers and Symbols

- Plaintext: AbC123!
- Shift Key:3
- XOR Key: 5

For the encryption,

Plaintext	Type	ASCII	Shifted ASCII (for only letters)	XOR Key	XOR Output	XOR Ciphertext
A	Letter	65	68(A-D)	5	68	D

b	Letter	98	101(be)	5	103	G
C	Letter	67	70(CF)	5	70	F
1	Number	49	-	5	52	4
2	Number	50	-	5	55	7
3	Number	51	-	5	54	6
!	Symbol	33	-	5	36	\$

Table 17: Encryption of Test 5

Ciphertext: DGF476\$

For the decryption,

- Ciphertext: DGF476\$
- Key: Shift by 3
- XOR key: 5

XOR Ciphertext	XOR Output	ASCII	Shifted ASCII(for letters)	Plaintext
D	65	65	65 (reverse Caesar)	A
G	98	98	98(reverse Caesar)	b
F	67	67	67 (reverse Caesar)	C

4	49	49	-	1
7	50	50	-	2
6	51	51	-	3
\$	33	33	-	!

Table 18: Decryption of Test 5

Plaintext: AbC123!

Test	5
Objective	To test the encryption and decryption of a complex string containing mixed case letters, numbers, and symbols.
Action	The plaintext AbC123! Is encrypted using a Caesar shift key of 3, followed by XOR encryption with key 5. Decryption is performed using XOR decryption and reverse Caesar shifting.
Conclusion	The original complex plaintext AbC123! is successfully reconstructed demonstrating the algorithm's versatility and reliability across different data types.

5. Evaluation

5.1 Strengths

- **Multi-Layer Security:** The algorithm integrates dynamic Caesar shifting, text reversal, and XOR encryption, creating layered protection that is significantly stronger than a traditional Caesar cipher.
- **Dynamic Shifting Mechanism:** The shift value varies with character position, reducing predictability and improving resistance against simple frequency and pattern-based attacks.
- **Bit-Level Protection:** XOR encryption operates at the ASCII level which is an effective way to mask data patterns and allows alphabetic and non-alphabetic characters.
- **Reliability and Reversibility:** The algorithm is symmetric, meaning the same keys are used for both encryption and decryption, ensuring reliable and consistent data recovery.
- **Flexibility:** Capable of encrypting different types of plaintext inputs, making it useful for experimentation and educational cryptography studies.
- **Academic Contribution:** Enlightens students on the possibility of improving classical encryption with modern logical operations to add more value to their learning.
- **Validated Performance:** Performance is also verified through correct and consistent scores when the test is applied in multiple instances of testing (ALPHABET, BRAVO, POOJA, BAUWA, SANDY).

5.2 Weaknesses

- **No Industry-Grade Security:** It is not an industry-grade level of security, though it is stronger than Caesar cipher.
- **Reducing Processing Overhead:** Multiple encryption layers add to computational work compared to a simple substitution cipher.
- **Important Dependency:** The security is fully dependent on the confidentiality of the shift and XOR keys, their disclosure will lead to the loss of the encryption power.
- **Scalability Concerns:** Could be slow when used in very large data or high-performance.
- **Complexity of Learning:** This process requires many steps, and a beginner might find it difficult to learn as compared to traditional Caesar encryption.

5.3 Potential Application Areas

1. Academic Learning and Coursework

Ideal for teaching core cryptographic principles such as substitution, bitwise operations, and reversible encryption.

2. Research and Experimental Cryptography

Useful for testing hybrid encryption models, evaluating layered security, and studying cipher behavior under different key conditions.

3. Prototype Cipher Design

It can be used as a foundation for developing advanced custom encryption schemes in educational or lab-based cryptography projects.

4. Low-Security Data Protection

Suitable for securing non-critical information such as student projects, internal notes, classroom demonstrations, and temporary data storage.

5. Secure Messaging in Controlled Environments

It Can be implemented in academic chat tools, puzzle platforms, or hobby projects where encryption is required but military-grade security is not.

Conclusion

The DLC-XOR algorithm has been able to expand the Classical Caesar Cipher to include dynamic shifting, reversal of text and encryptions based on XOR which reduces the predictability and low security of the original algorithm. Its accuracy, reliability and reversibility are proved by testing a variety of plaintexts and key values. DLC-XOR, though not a security tool to be used on an industrial level, can serve as a good educational and experimental tool to learn the use of classical and modern cryptography techniques in a specific instance.

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Its stratified design explains why conventional ciphers might be expanded to add a higher complexity, disguise patterns, and offer a flexible encryption of both alphabetic and non-alphabetic data. Overall, the algorithm fulfills its goals of providing a feasible example of hybrid encryption making contributions to research, experimentation, and learning in cryptography.

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